

基於深度學習與模糊PID控制之微型探針振動分散智慧閉迴路系統

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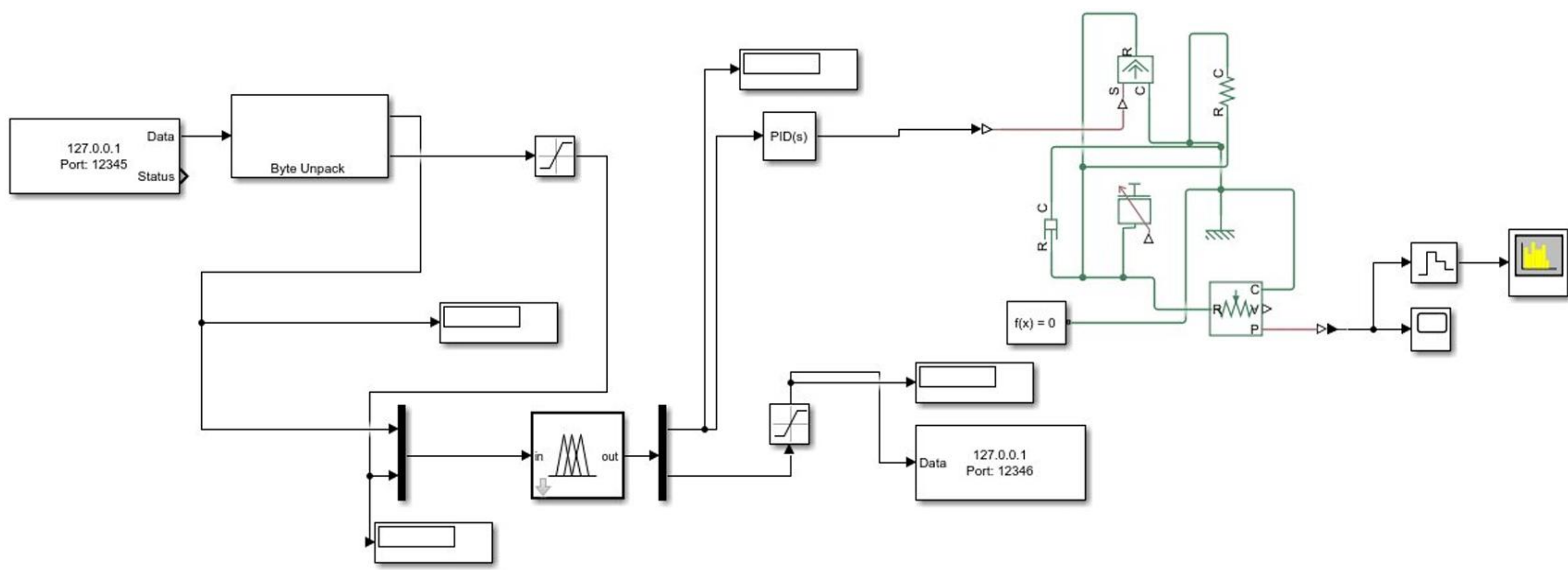
摘要

隨著半導體封測邁向微間距發展，探針卡針徑已微縮至 1.5–3.5 mil，表面並鍍有奈米級抗沾黏層。現行產線缺乏針對此類極輕、易勾連零件的自動化供料方案：傳統定頻振動盤易導致探針死結或彈跳受損，人工鑷子夾取則易刮傷奈米鍍膜。本研究提出整合電腦視覺與模糊PID 控制之智慧型閉迴路振動分散系統。系統以 YOLOv11 實例分割模型即時偵測探針分散狀態，計算均勻度（Uniformity）與數量比例（Single Ratio）兩項指標，經雙向 UDP 傳送至 MATLAB Simulink，由 Mamdani 型模糊推論系統（FIS）結合 PID 控制器進行閉迴路運算，輸出最佳振動頻率回饋至振動盤驅動器。

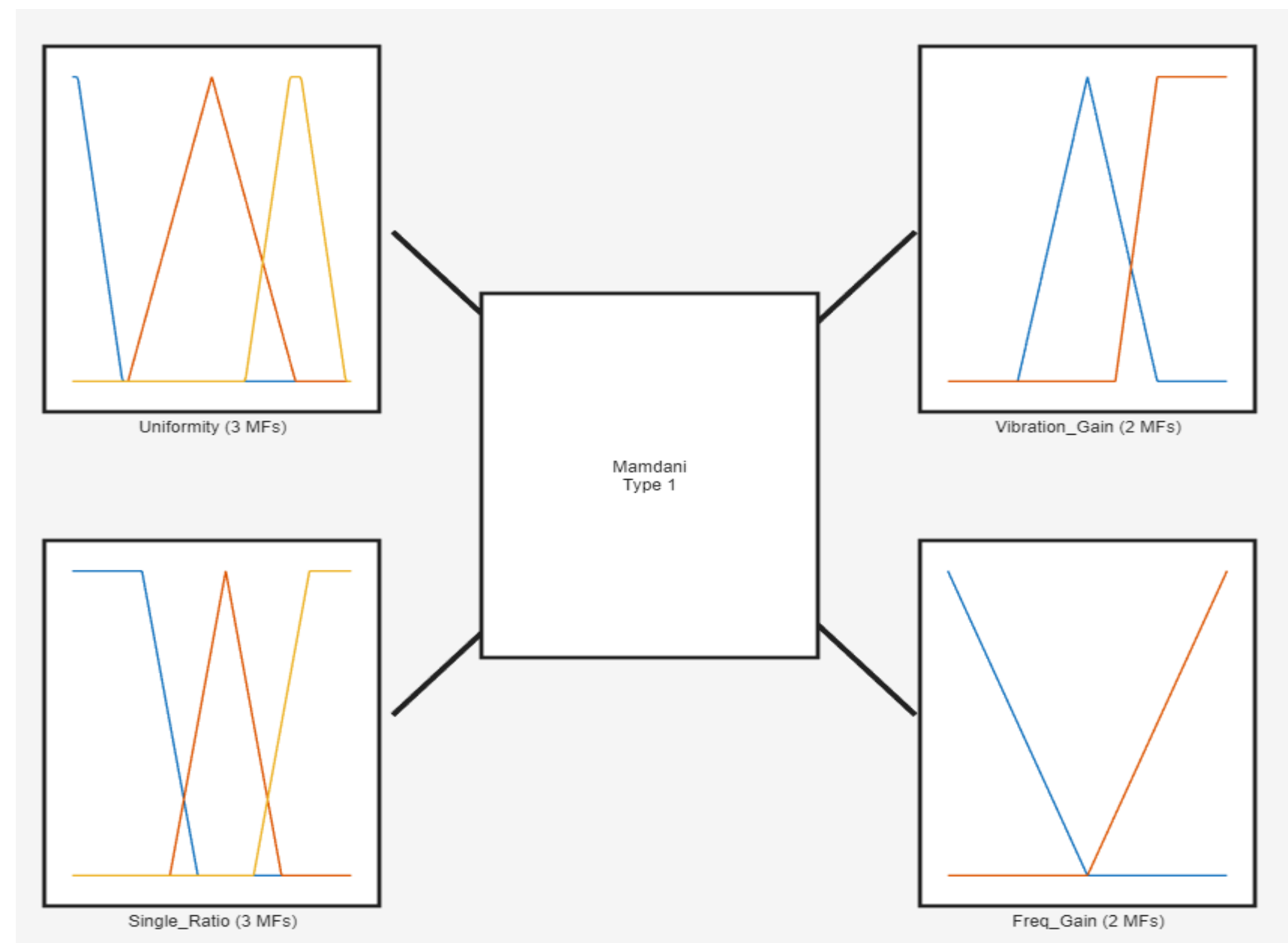
實驗架構



MATLAB Simulink 控制模型架構



Mamdani模糊推論

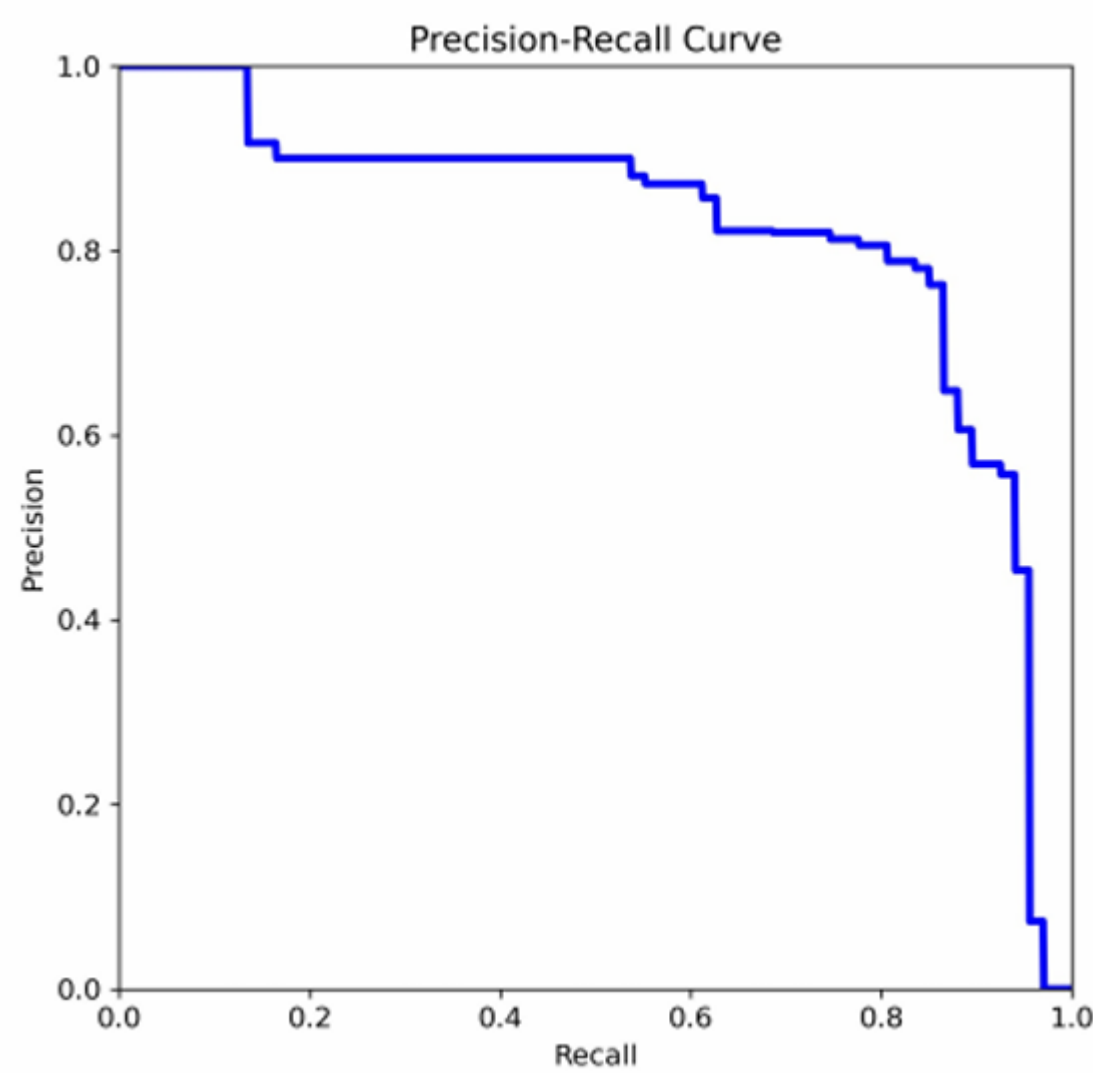


Mamdani FIS 規則表

	Rule	Weight	Name
1	If Uniformity is Poor and Single_Ratio is Few then Vibration_Gain is Impact, Freq_Gain is Impact	1	rule1
2	If Uniformity is Normal and Single_Ratio is Few then Vibration_Gain is Impact, Freq_Gain is Impact	1	rule2
3	If Uniformity is Good and Single_Ratio is Few then Vibration_Gain is Gentle, Freq_Gain is Gentle	1	rule3
4	If Uniformity is Poor and Single_Ratio is Normal then Vibration_Gain is Impact, Freq_Gain is Gentle	1	rule4
5	If Uniformity is Normal and Single_Ratio is Normal then Vibration_Gain is Gentle, Freq_Gain is Gentle	1	rule5
6	If Uniformity is Good and Single_Ratio is Normal then Vibration_Gain is Gentle, Freq_Gain is Gentle	1	rule6
7	If Uniformity is Poor and Single_Ratio is Many then Vibration_Gain is Gentle, Freq_Gain is Gentle	1	rule7
8	If Uniformity is Normal and Single_Ratio is Many then Vibration_Gain is Gentle, Freq_Gain is Gentle	1	rule8
9	If Uniformity is Good and Single_Ratio is Many then Vibration_Gain is Gentle, Freq_Gain is Gentle	1	rule9

影像辨識模型

- 模型：YOLOv11
- Epoch：100
- 解析度：640×640
- Precision：77.9%
- Recall：85.1%



閉迴路控制驗證結果

系統狀態	針數	均勻度	比例	PID 輸出	振動頻率	控制行為
正常分散	28	0.85	1.00	1.169	58.4 Hz	低頻維持
分散中	~10	0.50	0.35	1.674	83.7 Hz	中頻驅動
堆疊卡料	0	0.00	0.00	1.837	91.8 Hz	高頻衝擊

結果與未來展望

- 結果
- YOLOv11 + Simulink 模糊PID 整合，實現微型探針智慧閉迴路分散控制
 - 雙向 UDP 即時通訊，延遲 <1ms
 - Mamdani FIS 9 條規則，三種狀態完整驗證
 - 靜態圖片輪播機制，依頻率調整切換速度模擬閉迴路響應

- 未來展望
- 接入實體相機與振動盤，完成真實硬體閉迴路
 - DAQ / Arduino 輸出 PWM 驅動實體振動盤
 - 提升訓練解析度（640→1280），改善 Mask 精度